

A small icon of a flag with the number 7 on it.

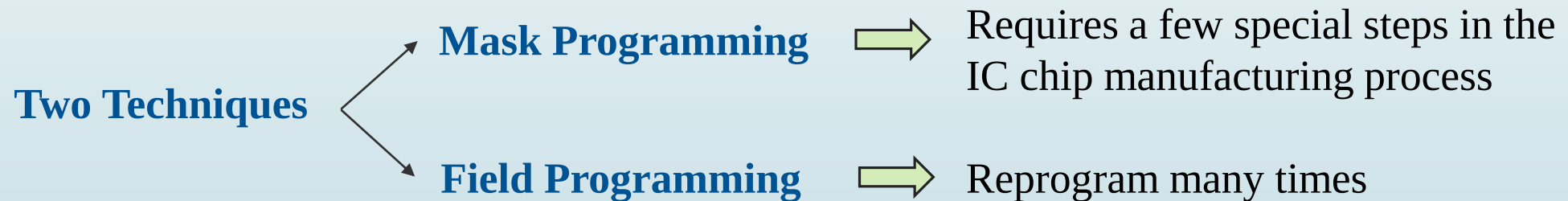
7. Programmable Logic Devices



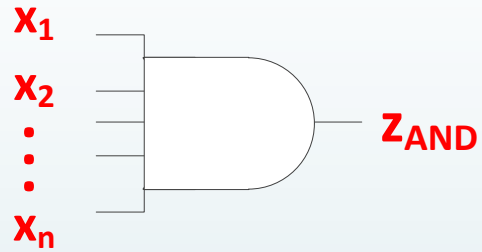
- 1. Programmable Arrays**
2. Field Programmable Gate Arrays (FPGA)
3. Register Level Design using FPGA cells

Programmable Logic Devices

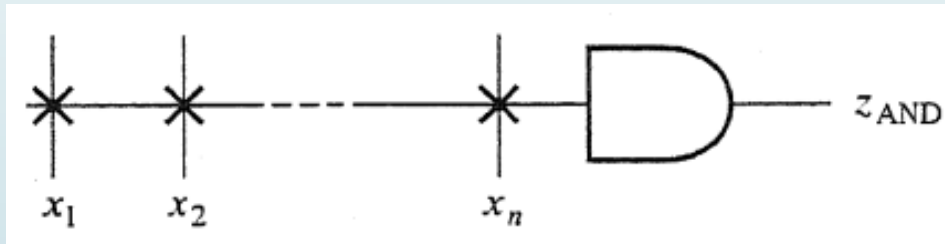
- contain many logic gates or other general purpose cells whose interconnections can be configured or “programmed” to implement any desired combinational or sequential function.



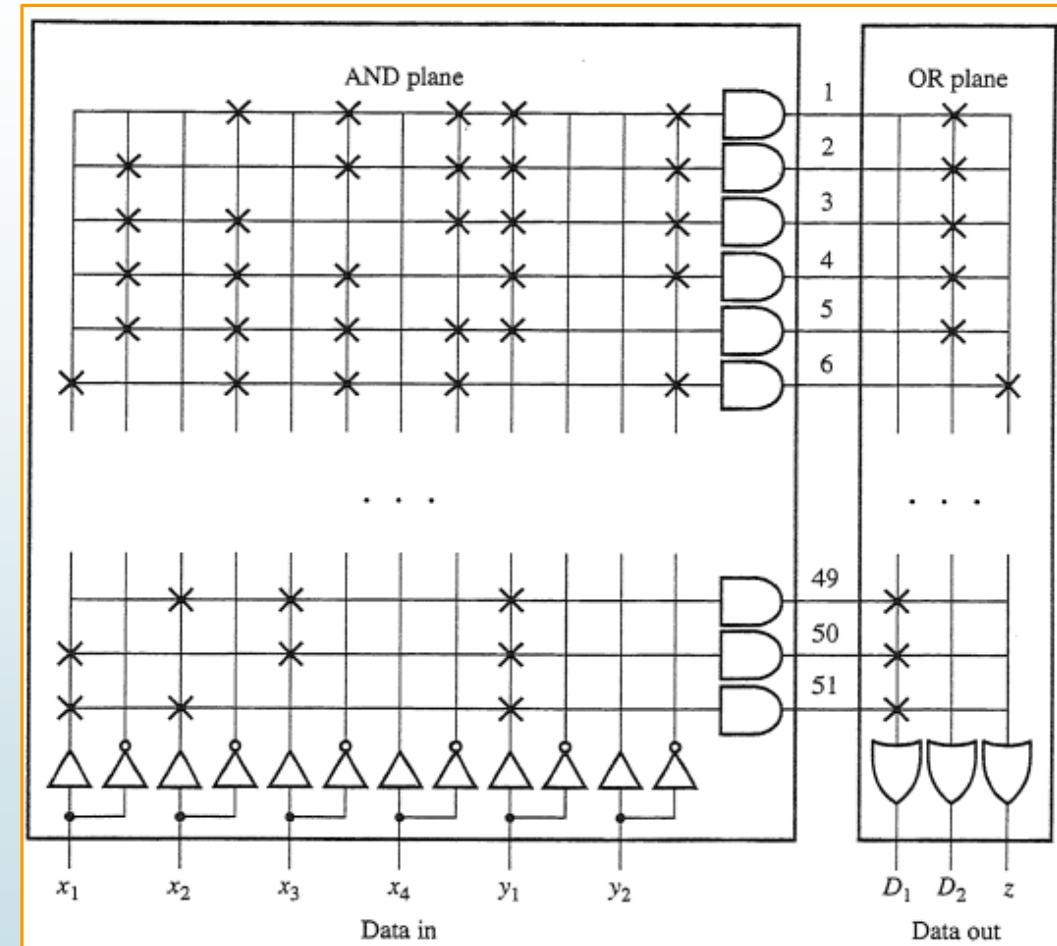
Programmable Array



n-inputs AND gate (normal notation)



n-inputs AND gate (PLD notation)

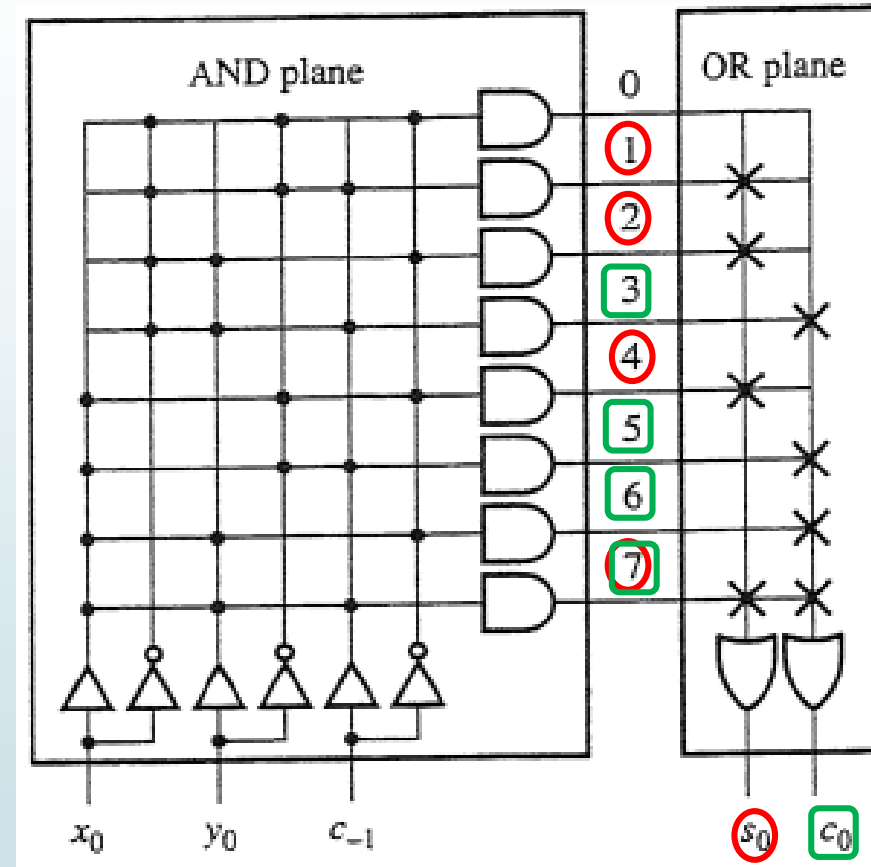


PLA implementing the Combinational Logic Circuit of 4-bit Adder

ROM implementation of a Full Adder

Inputs			Outputs	
x_0	y_0	c_{-1}	s_0	c_0
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Truth Table



ROM array



THANK YOU FOR YOUR ATTENTION